Garlock
SEALING TECHNOLOGIES®
an EnPro Industries company

Garlock 9920

Nuclear Grade Gasketing

MATERIAL PROPERTIES*

Color:	Mahogany
Composition:	Graphite fibers with a nitrile binder
Fluid Services[1]**:**	Saturated steam[3], water, oil, inert gases, aliphatic hydrocarbons and gasoline
Temperature[2], °F (°C)	
Minimum:	-100 (-73)
Continuous Max:	+650 (+343)
Maximum:	+1000 (+537)
Pressure[2], Maximum, psig (bar):	2000 (138)
P x T (max.)[2], psig x °F (bar x °C)	
1/32 and 1/16":	700,000 (25,000)
1/8":	350,000 (12,000)

PHYSICAL PROPERTIES*

ASTM F36	**Compressibility**, range, %:	7-17
ASTM F36	**Recovery**, %:	40
ASTM F38	**Creep Relaxation**, %:	20
ASTM F152	**Tensile**, Across Grain, psi (N/mm^2):	1000 (7)
ASTM F1315	**Density**, lbs./ft.^{3} (grams/cm^3):	90 (1.44)

IMMERSION PROPERTIES* - ASTM F146 Fluid Resistance after Five Hours

	ASTM #1 Oil 300°F (150°C)	ASTM IRM #903 300°F (150°C)	ASTM Fuel A 70-85°F (20-30°C)	ASTM Fuel B 70-85°F (20-30°C)
Thickness Increase, (%)	0-10	0-15	0-10	0-20
Weight Increase, (%)	<15	-	<15	<20
Tensile Loss, (%)	-	<60	-	-

Chemical Impurity Data*(5)

Total Heavy Metals					
Antimony, Max., ppm:	100	**Arsenic**, Max., ppm:	100	**Bismuth**, Max., ppm:	100
Cadmium, Max., ppm:	100	**Copper**, Max., ppm:	100	**Gallium**, Max., ppm:	100
Indium, Max., ppm:	100	**Lead**, Max., ppm:	100	**Mercury**, Max., ppm:	100
Tin, Max., ppm:	100	**Zinc**(4), Max., ppm:	10,000		
Leachable Levels					
Chlorides, Max., ppm:	200	**Fluorides**, Max., ppm:	100	**Sulfur**, Max., ppm:	1600

Notes:

This is a general guide and should not be the sole means of selecting or rejecting this material. ASTM test results in accordance with ASTM F-104; properties based on 1/32" (0.8mm) sheet thickness unless otherwise mentioned.

* Values do not constitute specification Limits

1 See Garlock chemical resistance guide.

2 Based on ANSI RF flanges at our preferred torque. When approaching maximum pressure, continuous operating temperature, minimum temperature or 50% of maximum PxT, consult Garlock Applications Engineering. Minimum temperature rating is conservative.

3 Minimum recommended assembly stress = 4,800psi. Preferred assembly stress = 6,000-10,000psi. Gasket thickness of 1/16" strongly preferred. Retorque the bolts/studs prior to pressurizing the assembly. For saturated steam above 150psig or superheated steam, consult Garlock Engineering.

4 This material shows a high level of zinc in the form of zinc oxide.

5 The chemical impurity data and physical test results, certification to both 10CFR.50 Appendix B and 10CFR.21, will always be included with the product.